

Venue and dataset decision for September 29, 2026

Prepared September 28, 2026. The authors confirmed that both previous submissions were rejected or withdrawn. Recommendation: target **ISBI 2027, October 26, 2026**, if the evidence and reproducibility work below can be completed. The conference deadline is approximately four weeks away; acceptance and the conference itself occur later.

Conference shortlist

Priority	Venue	Official submission deadline	Fit and recommendation
1	IEEE ISBI 2027	October 26, 2026	Best match to the requested one-month window and biological image analysis. Focus on a compact, reproducible 2D study with external boundary evaluation.
2	MIDL 2027	Abstract November 30; full paper December 4, 2026, 23:59 AoE	Strong fit for deep learning and rigorous external validation; allows more time to complete adaptation, ablations, and source provenance.
3	IPMI 2027	Abstract December 7; full article December 14, 2026	Strong imaging-methods audience, but a substantially stronger methodological contribution would be needed for this project.
4	MICCAI 2027	Not yet announced on the MICCAI Society page checked today	Strong future target for a mature method; not a verified one-month submission option.

Official sources: [ISBI instructions](#), [MIDL dates](#), [MIDL scope](#), [IPMI dates](#), [MICCAI Society](#).

ISBI limits technical material to four pages. A paid fifth page can contain references, acknowledgments, and ethical compliance information only. Its one-page abstract track is not included in the proceedings. The current PDF is a readable meeting draft, not the final ISBI template. Recheck the deadline time zone and submission portal when it opens; the checked author page states the date but not a time zone.

MIDL is the best alternative if the October deadline would force an incomplete experiment. A validation-focused framing is more credible than claiming that the present architectural additions outperform all alternatives. The clinical-translation track should only be chosen if the finished study satisfies its biomedical/operational validation requirements; cultured-cell experiments are not clinical validation. Avoid overlapping submissions while a manuscript is under review.

Journals

The three venues mentioned in the request are journals, not conferences. Standard research-article submission is not tied to the conference dates above. Their fit is more useful here than a prestige ranking:

1. **Biomedical Signal Processing and Control**: strongest match among the three for image preprocessing, Gabor features, optimization, and boundary evaluation. It still needs a measurable methodological contribution and adequate validation. [Publisher scope](#). 2. **Computers in Biology and Medicine**: good target if the paper establishes biological utility, robust dataset partitions, and meaningful downstream morphometry. Its publisher explicitly flags unclear splits and imprecise segmentation-task analysis as unsuitable. [Publisher scope](#). 3. **Expert Systems with Applications**: consider after demonstrating a broadly useful AI method across several domains; a breast-only architecture combination with no consistent gain is a weaker fit. [Publisher scope](#).

These are judgments about scope and readiness, not acceptance forecasts. The existing evidence does not support a state-of-the-art or clinical-deployment claim.

Dataset choices

Dataset	What is labeled	Use in this package	Role in the paper
BBBC038v1 / DSB2018	Individual nuclear masks across varied light microscopy	All 65 official stage-1 test images, original released RLE annotations	Main appearance-shift stress test; nucleus-mask union scored at native resolution
BBBC039v1	Hoechst-stained U2OS nuclei, 16-bit fluorescence	All 50 official test images; 100 train and 50 validation images remain unused here	Closest tested image/label match to fluorescence nucleus segmentation; best next controlled adaptation benchmark
TNBC v1.1	Annotated nuclei in breast histology; 50 images from 11 patients	All 50 images; patient-level summaries and paired tests	Breast-specific, larger modality shift; current failure is useful evidence, not a successful transfer claim
LIVECell	Whole-cell instances in phase-contrast imaging, including MCF7, BT474, SkBr3	Researched, not downloaded or evaluated	Suitable only if the target is expanded from nuclei to whole cells; well/location/time identifiers support grouped splits
CryoPPP	Protein-particle coordinates in cryo-EM micrographs	Researched, not evaluated	A separate particle-detection task; coordinates do not establish accurate object contours
CryoNuSeg	Nuclear masks in cryosectioned H&E histology	Researched, not evaluated	Optional future histology test; cryosectioned histology is not cryo-EM

Dataset sources and terms: [BBBC038](#), [CC0](#), [BBBC039](#), [CC0](#), [TNBC v1.1](#), [CC BY 4.0](#), [LIVECell](#), [CC BY-NC 4.0](#), [CryoPPP authors' repository](#), [CryoNuSeg original paper](#).

BBBC039's authors explicitly report overlap with BBBC038. These two benchmark results must remain separate; they cannot be pooled as independent cohorts. Before any cross-dataset training, audit overlap at source/patch level, including resized and contrast-transformed duplicates. Hash matching alone cannot exclude these. No public benchmark data were used for training or parameter selection in the current runs. Public-pretraining exposure of Cellpose-SAM has not been audited here, so its results are an off-the-shelf reference rather than proof of unseen-domain generalization under matched data access.

For cryo-EM, use coordinate-matching precision/recall and localization error with a stated tolerance, and split by EMPIAR experiment/protein. Synthetic disks around coordinates are weak labels, not independently annotated segmentation truth. This direction would distract from a four-page nucleus-segmentation submission and is deferred.

Four-week work plan

Dates	Work	Deliverable and decision criterion
September 29-October 2	Agree the scientific claim; recover acquisition IDs; document annotation protocol; lock public train/validation/test records	Source-level manifest and an honest claim that all authors endorse
October 3-9	Compare the revised model with vanilla B7 UNet++, smaller U-Net, and Cellpose-SAM; use BBBC039 official train/validation only for adaptation, and patient-grouped development for TNBC	Three seeds, equal learned-model budgets, learning curves, frozen validation selection; clearly mark exploratory use of benchmarks already inspected
October 10-15	Isolate normalization, edge channel, polarity handling, and optional boundary/distance supervision; evaluate all planned ablations	Improvement must survive boundary metrics and appropriate paired source-level uncertainty; keep negative results
October 16-19	Evaluate locked choices on an additional previously uninspected cohort; obtain repeat annotation of a predefined private subset	Independent confirmation plus inter-annotator Dice/surface distances; no invented annotation ceiling
October 20-23	Assemble four-page ISBI paper and public release	Actual figures, accessible weights, exact environment, raw result tables, checked claims and references
October 24-26	Professor/coauthor review, template and portal checks, final submission	All authors approve; switch to MIDL if core evidence or release terms remain unresolved

The highest-priority methodological experiment is **controlled domain adaptation and intensity normalization**, because the new measurements show a large imaging-modality dependence. A distance or boundary auxiliary head is the next candidate, but its value must be isolated against plain Dice+BCE and the same encoder. Increasing network size is not supported by the current ablation evidence.

Decisions for the professor: approve the replacement-paper framing; choose ISBI versus the longer MIDL schedule; confirm stain/acquisition/annotation details; nominate a second annotator; approve private-data/weight release terms. Keep 3D to one short subsection or supplementary figure, describing selected-plane foreground evaluation rather than dense 3D instance accuracy.